

Leblanc, Benjamin

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EDUCATION

Doctor of Philosophy in Computer Science (Ph.D.), with honors University Laval, Quebec	2022-2026
Certificate in Philosophy University Laval, Quebec	2021-2023
Master of Science in Computer Science, with thesis (M.Sc.) University Laval, Quebec	2020-2021
Bachelor of Science in Actuarial Science, international profile (B.Sc.) University Laval, Quebec	2017-2020

WORK EXPERIENCE

University Instructor (University Laval)	2023-2024
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Responsible for teaching the course Mathematics for Computer Scientists (MAT-1919) for the Fall 2023 session (in-person) and Winter 2024 session (co-modal). Creation of exams, assignments, and quizzes. Course curriculum: "Sets, relations, recurrences, trees and graphs. Theory of finite and infinite sets, notions of functions, generating series and recurrence resolutions, counting principles and combinatorial analysis, elements of graph theory and trees. Notions of mathematical proofs, notably proofs by induction and by contradiction. Review of limits, integrals, sequences, series and linear system resolutions."

Business Intelligence Intern (Co-operators Insurance and Financial Services)	2019-2020
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Participated in the development and optimization of a tool for pricing in property and casualty insurance. This tool primarily serves for sorting and processing data concerning policies, insureds, and what the policy covers in order to obtain various statistics on filtered criteria. Participated in the maintenance of various databases as well as the transfer of history from an obsolete database to a more recent one.

PUBLICATIONS IN CONFERENCES AND JOURNALS

- **Leblanc B.** and Germain P. A Framework for Bounding Deterministic Risk with PAC-Bayes: Applications to Majority Votes. **TMLR (submitted)**, 2026. [[preprint](#)]
- Bastian J., **Leblanc B.**, Germain P., Habrard A., Largeron C., Metzler G., Morvant E., and Viallard P. PAC-Bayesian Generalization Guarantees for Fairness on Stochastic and Deterministic Classifiers. **NeurIPS (submitted)**, 2026. [[preprint](#)]
- Bastian J., **Leblanc B.**, Germain P., Habrard A., Metzler G., Morvant E., and Viallard P. On the Disintegration of the Stochastic Majority Vote: From PAC-Bayesian Bounds to a Self-Bounding Algorithm. **TMLR (submitted)**, 2026. [[preprint](#)]
- **Leblanc B.**, Bazinet M., D'Amours N., Drouin A., and Germain P. Generalization Bounds via Meta-Learned Model Representations: PAC-Bayes and Sample Compression Hypernetworks. **ICML (main track)**, 2025. [[publication](#)][[preprint](#)]
- **Leblanc B.** and Germain P. Seeking interpretability and explainability in binary activated neural networks. **XAI World Conference (main track)**, 2024. [[publication](#)][[preprint](#)]
- Sanchez-Reinoso Z., Bazinet M., **Leblanc B.**, Clément J.-P., Germain P., and Bazinet L. Application of machine learning tools to study the synergistic impact of physicochemical properties of peptides and filtration membranes on peptide migration during electrodialysis with filtration membranes. **Journal of Separation and Purification Technology**, 2024. [[publication](#)]
- Fortier-Dubois L., **Leblanc B.**, Letarte G., Germain P., and Laviolette F. PAC-Bayesian Learning of Aggregated Binary Activated Neural Networks with Probabilities over Representations. **Canadian AI (main track)**, 2023. [[publication](#)]

PUBLICATIONS AT WORKSHOPS

- **Leblanc B.**, Lebel L.-J., Kana T., and Kamel R. Simplicity Suffices for Parameter Noise Injection in Stochastic Gradient Descent. **IJCAI (Data Science Meets Optimisation (DSO) Workshop)**, 2026. [[preprint](#)]
- **Leblanc B.**, Bazinet M., D'Amours N., Drouin A., and Germain P. Sample Compression Hypernetworks: From Generalization Bounds to Meta-Learning. **NeurIPS (Neural Compression Workshop)**, 2024. [[publication](#)][[preprint](#)]
- **Leblanc B.** and Germain P. Seeking interpretability and explainability in binary activated neural networks. **IJCAI (Workshop on Explainable Artificial Intelligence (XAI))**, 2023. [[publication](#)]
- **Leblanc B.** and Germain P. A Greedy Algorithm For Building Compact Binary Activated Neural Networks. **Montreal AI Symposium**, 2022. [[publication](#)][[preprint](#)]
- Fortier-Dubois L., **Leblanc B.**, Letarte G., Germain P., and Laviolette F. Learning Aggregations of Binary Activated Neural Networks with Probabilities over Representations. **Montreal AI Symposium**, 2021. [[publication](#)]

GRANTS AND AWARDS

- Doctoral thesis awarded honors (Université Laval, 2026).
- Recipient of the François-Laviolette Scholarship for the development of responsible AI.
- Recipient of the Pierre-Ardouin Prize for the year 2022-2023, rewarding the best session work, all graduate or multi-cycle courses combined, within the Department of Computer Science and Software Engineering at University Laval.
- Recipient of the best presentation award at NumériQC Week 2023 - AI and Data Rising Stars Competition.

MENTORING

- Louis-Jacob Lebel (Bachelor's, University Laval, 2025-2026)
- Richard Kamel (Bachelor's, University Laval, 2025-2026)
- Teddy Kana (Bachelor's, University Laval, 2025-2026)

ACADEMIC SERVICE

- **Reviewer**
 - o NeurIPS (2023, 2026)
 - o ICML (2023, 2024, 2025)
 - o ICLR (2024, 2025)
 - o IJCAI (2024-25)
 - o CHI (2024)
 - o ANNPR (2024)
 - o NLDL (2025)
- **Social involvement**
 - o Collaborator (regular guest) on the [I.A. Café podcast](#) (AI vulgarization).
- **Teaching (french)**
 - o [Mathematics for Computer Scientists](#) (2023-2024); University Laval, Quebec